

Camera-Based Weightlifting/Exercise Form Analysis and Feedback System

Project Proposal for UCS503
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1 Higher-order goal

... secondary framing

This project aims to improve individual training safety and consistency by giving fitness practitioners automated, camera-based feedback on their weightlifting technique. While the broader motivation is accessible, coach-free technique guidance, the immediate engineering objective is to translate *pose estimation and geometric biomechanical analysis* into a measurable, deployable software solution that runs on hardware most users already own – a laptop webcam.

2 Time-to-value

... primary heuristic

- We will deliver a working, testable increment quickly by building one exercise (Squat) fully end-to-end – pose estimation, rep detection, form rules, and feedback – before extending to Deadlift and Bench Press.
- We will prioritize a verifiable feedback loop over exhaustive exercise coverage, since a single reliable exercise pipeline validates the full architecture and de-risks the remaining two.
- Success in the first iteration is defined as: correct rep counting and at least one correctly detected form violation on self-recorded Squat footage, measured against manual review.

3 Problem Statement

Recreational and intermediate lifters who train without a coach or training partner have no objective, repeatable way to assess their own exercise technique. Common pain points include:

- **No real-time awareness:** a lifter cannot see their own knee alignment, depth, or back angle while performing a lift.
- **Manual video review is slow:** filming and reviewing sets after the fact is time-consuming and requires technique knowledge to interpret correctly.
- **No structured history:** without dedicated tooling, there is no consistent record of technique quality across sessions, so progress (or regression) goes unnoticed.

- **Hardware barrier:** wearable-sensor or motion-capture-based solutions require equipment beyond what a typical home or gym user owns.

Why this matters now (contextual relevance): lightweight, real-time pose-estimation models (e.g., MediaPipe Pose) now run efficiently on consumer laptops without specialized hardware, making a camera-only technique-feedback tool practically buildable within a single-semester timeframe.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a **browser-based** “Form Capture-to-Feedback” system with the following components:

- **Exercise selection & camera capture:** user selects Squat, Deadlift, or Bench Press and grants webcam access.
- **Real-time pose pipeline:** pose estimation, landmark smoothing, and joint-angle calculation running client-side.
- **Rep detection & form analysis:** a per-exercise movement state machine counts repetitions; a geometric rule engine flags common technique issues per rep.
- **Feedback & scoring:** prioritized real-time text feedback during the set, plus a post-session form score and summary.
- **History dashboard:** session-level metrics (not video) are stored and displayed as trends over time.

4.2 Core workflow

1. User selects an exercise and starts a session after a camera visibility check.
2. System tracks pose per frame, verifies the movement matches the selected exercise, and advances a rep-detection state machine.
3. On each completed rep, form rules evaluate the rep’s angle history and the top-priority issue (if any) is shown to the user.
4. On session end, aggregate metrics and a form score are computed and saved; the dashboard is updated.

4.3 Operational constraints for an educational, single-semester setting

- Keep the solution usable on a standard laptop webcam; no specialized sensors, depth cameras, or GPUs required.
- Avoid dependence on a large proprietary or novel ML training pipeline; focus on pose landmarks, geometry, and rule correctness.
- Design for deployability using common web hosting and a managed relational database.

5 Solution Approach

... Engineering focus

This is an engineering course project, so we focus on a correct, modular real-time pipeline rather than novel computer-vision research.

5.1 Form analysis strategy

... non-research

Form analysis will be rule-based, not classifier-based:

- Extract exercise-specific joint angles (e.g., knee, hip, torso for Squat) from smoothed pose landmarks.
- Compare each rep's angle trajectory against configurable thresholds (e.g., minimum depth, maximum knee drift) without claiming a validated clinical or biomechanical model.
- Surface only the single highest-priority violation per rep to the user, never an unranked list of warnings.

5.2 Web architecture

... browser-based solution

A client-heavy three-tier architecture:

- **Frontend (client):** React UI running camera capture, MediaPipe Pose, rep detection, form analysis, and feedback in real time.
- **Backend API:** FastAPI service for authentication, session submission, and history/dashboard queries.
- **Database:** PostgreSQL storing structured session, rep, and form-issue records – never workout video.

5.3 CI/CD and fast delivery enabler

A CI/CD pipeline will be used to:

- Automatically run tests (angle calculation, state-machine logic) on every change.
- Produce repeatable deployment artifacts for the frontend and backend.
- Enable safe, frequent releases as each exercise (Squat, then Deadlift, then Bench Press) is completed.

6 Evaluation Criterion

... measurable and attributable

We define success using measurable metrics tied to the core pipeline.

6.1 Primary evaluation metric

Rep-counting accuracy: agreement between system-counted repetitions and manually counted repetitions on self-recorded test sets.

- Target: high agreement across a self-recorded test set covering all three exercises under normal camera conditions.
- Attribution: measured by comparing logged rep counts against a human reviewer's manual count on the same recordings.

6.2 Secondary evaluation metrics

- **Form-detection agreement:** percentage of deliberately introduced technique errors (e.g., shallow squat, rounded back) that are correctly flagged by the rule engine.
- **Visibility-handling correctness:** whether the system correctly pauses/warns rather than emits confident feedback under poor lighting or occlusion.
- **Session completeness:** percentage of sessions that save successfully to the database without data loss.
- **Reliability:** basic uptime/availability of the staging/production deployment during testing windows.

6.3 Pilot validation plan

... high-level

- Record a self-collected test set (self and, where possible, a small number of additional volunteers) performing clean and deliberately flawed reps for each exercise.
- Compare system output (rep counts, flagged issues) against manual review of the same footage.
- Where accessible, cross-check Squat detections against a public labelled dataset (e.g., Fitness-AQA) as an external reference point.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** the system should support growth in users and sessions by:
 - keeping real-time pose processing client-side, so backend load scales only with session-submission and dashboard traffic,
 - decoupling frontend and backend,
 - using indexed queries for session/history lookups.
2. **Operationally deployable (practically):** the system should run on common web hosting:
 - managed PostgreSQL database,
 - standard frontend hosting for the React app,
 - CI/CD pipeline for staging/production releases.

8 Engine availability heuristic

A mature set of “engines” (tooling/libraries) is available to implement this as a browser-based project:

- MediaPipe Pose (Tasks API, JavaScript) for real-time, on-device pose landmark extraction.
- Web framework for REST APIs (FastAPI).
- Managed relational database (PostgreSQL).
- Authentication approach (token-based).

- Charting library for the React dashboard.
- Test framework for unit/integration tests (state machine, angle math, API endpoints).
- CI runner and staging deployment target.

We will use these mature, existing components to focus on the pipeline design, rule correctness, and measurement, rather than inventing new computer-vision infrastructure.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

- Camera capture + pose landmark extraction with live overlay
- Squat rep-detection state machine with smoothing/hysteresis
- Squat form rules (depth, knee alignment, torso angle) with prioritized real-time feedback
- Session save to backend (rep count, metrics, issues – no video)
- CI: build + unit tests for angle/state-machine logic

9.2 Subsequent deliverables

- Deadlift and Bench Press state machines and form rules
- Configurable weighted form score
- History dashboard with session trends and common-issue summaries
- Calibration against a self-recorded test set (and, where accessible, an external dataset reference check)
- CD to staging with a repeatable deployment pipeline

10 Risks and mitigations

- **Single-camera accuracy limits:** depth-dependent metrics (e.g., precise bar path) cannot be reliably measured from one webcam; such metrics will be explicitly labelled as approximate or unavailable rather than presented as accurate.
- **Noisy pose data / false rep counts:** mitigated via landmark smoothing and hysteresis thresholds in the state machine.
- **No large labelled weightlifting-form dataset:** V1 does not depend on one – form analysis is rule-based; a self-recorded test set is used for calibration and evaluation instead.
- **Scope creep beyond three exercises:** mitigated by building and validating Squat completely before starting Deadlift or Bench Press (see Section 2).
- **Privacy concerns:** mitigated by never storing workout video – only structured, derived metrics are persisted.

11 Summary

This project proposes a deployable, browser-based system that gives weightlifters automated, real-time technique feedback for Squat, Deadlift, and Bench Press using only a standard laptop webcam. It meets the course criteria by emphasizing time-to-value through an incremental, one-exercise-at-a-time build order, using measurable evaluation criteria (rep-counting accuracy and form-detection agreement against manual review), and remaining focused on engineering correctness and modular pipeline design rather than novel computer-vision research or dependence on a large proprietary dataset.